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Deep Learning for Used Car Price Prediction: A Computer Vision Approach

Case Study: Biskra Automotive Market

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Abstract

The used car market in the Biskra region of Algeria relies heavily on subjective pricing, with no standardized valuation system. This project proposes an automated deep learning pipeline that estimates used car prices directly from visual data. By combining **YOLOv5** for vehicle detection and **ResNet-18** for feature extraction, we built an end-to-end regression system trained on a locally compiled dataset of **281 car images** from the Biskra automotive market, each labeled with its real market price (in million DZD), mileage, and body condition. The entire system was developed and deployed locally using Python, PyTorch, and a Streamlit web interface — without reliance on cloud platforms. The model demonstrates promising predictive performance and represents a first step toward a fair and transparent pricing tool for the local automotive market.

Keywords: Deep Learning, Computer Vision, YOLOv5, ResNet-18, Price Prediction, Used Cars, Biskra, Algeria, PyTorch, Streamlit.

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Chapter 1

Introduction

1.1 Context and Motivation

The twenty-first century is witnessing an unprecedented technological transformation driven by Artificial Intelligence (AI). This revolution is reshaping industries, economies, and daily life. One of the most dynamic sub-fields of AI is **Computer Vision**, which gives machines the ability to see and interpret visual information. Its applications now extend far beyond self-driving vehicles, reaching logistics, insurance, and market valuation.

In the local context of Algeria, specifically in the **Biskra** region, the used car market is highly active but lacks any standardized pricing mechanism. Sellers and buyers depend on personal estimation, market experience, and word of mouth — a process that is slow, inaccurate, and often unfair.

This project aims to bridge this gap using state-of-the-art deep learning to automate the price estimation process. By analyzing photos of vehicles, our system provides an objective market value estimate, reducing the time and effort required for manual inspection.

1.2 Problem Statement

The core problem addressed by this project is: *Can a machine learning model reliably predict the market price of a used car in Biskra from a single photograph?*

This problem is non-trivial because:

- Car prices depend on multiple visual factors: brand, condition, visible damage, and model year.
- The Biskra market has its own local price dynamics, different from European or American datasets.
- No existing labeled dataset targets the Algerian local car market.

1.3 Objectives

The specific objectives of this work are:

1. To build a locally-sourced, labeled image dataset from the Biskra automotive market.
2. To design and implement a deep learning regression pipeline combining YOLOv5 and ResNet-18.
3. To develop a functional prototype web application for real-time price estimation.
4. To evaluate the model's performance using standard regression metrics.

1.4 Report Structure

This report is organized as follows: Chapter 2 reviews the relevant literature. Chapter 3 presents our methodology. Chapter 4 details the implementation. Chapter 5 presents results and evaluation. Chapter 6 concludes the work.

Chapter 2

State of the Art

"... the foundational principles of neural networks are well-documented in [1]."

2.1 Evolution of Deep Learning in Computer Vision

Computer Vision has undergone a radical transformation with the rise of deep learning. Before CNNs (Convolutional Neural Networks), researchers relied on hand-crafted features such as SIFT and HOG, which generalize poorly across diverse visual conditions. CNNs enabled models to learn hierarchical features directly from raw image pixels, achieving breakthrough results in classification, detection, and regression tasks.

2.2 Key Architectures Used in This Project

"... YOLO treats object detection as a single regression problem [4]."

2.2.1 YOLOv5: Real-Time Object Detection

YOLO (You Only Look Once) [6] treats object detection as a single regression problem — directly predicting bounding boxes and class probabilities from full images in a single forward pass. This gives YOLO a major speed advantage over region-proposal methods such as R-CNN.

In our project, **YOLOv5** (implemented via `torch.hub`) is used to:

- Locate the vehicle within the input image using a bounding box.
- Crop the car region, eliminating irrelevant background (buildings, people, sky).

This preprocessing step is critical for ensuring that the subsequent price estimator focuses only on the vehicle.

"... ResNet-18, a lightweight variant, was introduced by He et al. [2]."

2.2.2 ResNet-18: Feature Extraction for Regression

ResNet (Residual Network) [**?**], introduced by He et al. (2016), solved the vanishing gradient problem in deep networks by introducing skip connections that allow gradients

to flow directly through the network. We use **ResNet-18**, a lightweight but powerful variant, as our feature extractor. Its final fully connected layer is replaced by a single-output regression head trained to predict car prices.

2.3 Why This Architecture Combination?

The two-stage pipeline (YOLO for detection + ResNet for regression) is justified by:

- **Separation of concerns:** YOLO handles localization; ResNet handles price estimation.
- **Computational efficiency:** Both models run on standard consumer hardware without requiring GPU cloud services.
- **Modularity:** Each component can be improved independently.

2.4 Related Works

Several prior works inform our approach:

1. **Stanford Car Dataset:** Used for 196-class car model classification [**?**]. Our work goes further by predicting continuous price values (regression).
2. **Carvana Image Masking (Kaggle):** Focused on vehicle background removal — similar to our YOLO-based cropping step.
3. **Automatic Vehicle Damage Assessment:** Insurance-industry AI systems estimate repair costs from exterior photos — the same visual logic we apply to total market valuation.
4. **Autobip.com:** An Algerian platform tracking local market prices, used in our analysis of price trends for SEAT Ibiza and Kia Picanto.

Chapter 3

Methodology

3.1 Overview of the Pipeline

Our system transforms a raw car photograph into a price estimate through the following sequential steps:

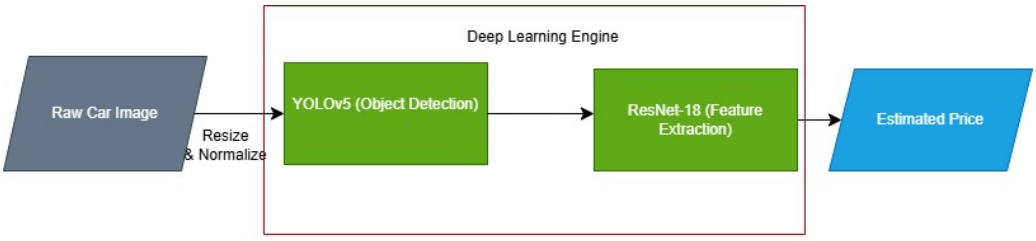

Figure 3.1: System Pipeline: From Input Image to Price Prediction.

The proposed system follows a modular architecture as illustrated in Figure 3.1. The raw input image undergoes initial detection via YOLOv5, which isolates the vehicle from complex backgrounds. The cropped vehicle image is then processed by a ResNet-18 model, modified to function as a regressor, to compute the final market price estimate.

3.2 Phase 1: Data Acquisition from Biskra Market

All data was collected directly from the local used car market in Biskra. The collection process involved:

- Photographing vehicles at the market, capturing one or more images per car.
- Recording the actual asking price, mileage, and body condition for each car.

Figure 3.2: Sample Images from the Biskra Local Car Market Dataset.

3.3 Phase 2: Dataset Construction and Labeling

Raw images were organized and labeled by **Mamen Mohamed Abdelghani** and **Mohamed Youcef** using Microsoft Excel.

1. **Image Cleaning:** Blurry, duplicated, or irrelevant images were removed manually. A Python script (`fix_images.py`) was written to detect and correct naming errors (e.g., double extensions such as `.jpg.jpg`).

2. **Labeling:** Each image was assigned three labels: price (in million DZD), mileage (in km), and body condition (0 = good, 1 = used/worn).

3. **Data Merging:** Multiple Excel sheets from different collection sessions were cleaned and merged using `clean_data.py` and `Pandas` into a unified `combined_data.csv`.

3.3.1 Dataset Statistics

Table 3.1: Summary Statistics of the Biskra Car Market Dataset.

Property	**Value**
Total images labeled	281 (target: 400)
Price range	100 – 500 million DZD
Average price	≈ 194 million DZD
Average mileage	≈ 136,000 km
Body condition: Good (0)	80 samples (34%)
Body condition: Worn (1)	157 samples (66%)
Image resolution	1080×606 to 1280×720 px

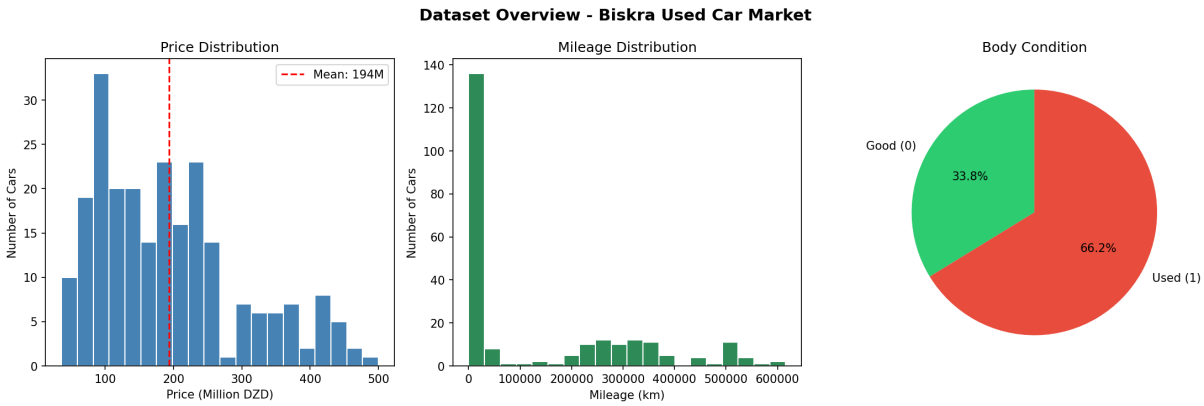

Figure 3.3: Dataset Overview: Price Distribution, Mileage Distribution, and Body Condition.

3.4 Phase 3: Model Architecture

The deep learning pipeline is implemented entirely in **PyTorch** and consists of two stages:

1. **Stage 1 – Vehicle Detection (YOLOv5):** The input image is passed to YOLOv5s (loaded via `torch.hub`). The model detects all objects in the image, and we extract the bounding box of the first detected "car" object. The image is then cropped to this region.

2. **Stage 2 – Price Regression (ResNet-18):** The cropped vehicle image is resized to 224×224 pixels and normalized with ImageNet statistics. It is then passed through **ResNet-18**, whose final fully connected layer (`fc`) has been replaced by a single-neuron linear layer to produce a scalar price prediction.

The trained model weights are saved as `car_price_model.pth` and loaded at inference time.

Chapter 4

Implementation

"... the pipeline is implemented using PyTorch [5]."

4.1 Development Environment

The entire project was developed and runs **locally on a personal computer** using the Windows operating system. All dependencies were installed manually via the terminal (Command Prompt / pip). No cloud platform was used.

Table 4.1: Software Stack Used in the Project.

Tool / Library	**Role**
Python 3.10	Programming language
PyTorch + torchvision	Model definition, training, inference
YOLOv5 (torch.hub)	Vehicle detection
ResNet-18 (torchvision)	Feature extraction + regression
OpenCV (cv2)	Image reading and preprocessing
Pandas & NumPy	Dataset management and CSV processing
Streamlit	Web-based user interface
Matplotlib	Training visualization
Microsoft Excel	Manual data labeling
Git / GitHub	Version control

4.2 Data Processing Scripts

4.2.1 `clean_data.py` – Data Extraction and Cleaning

This script reads the raw Excel file produced during manual labeling. It uses regular expressions to extract image IDs, prices, mileage, and body condition from each row. Rows with missing prices are dropped. The output is a clean `data.csv` ready for model training.

4.2.2 `fix_images.py` – Image Name Correction

During dataset construction, some images were accidentally saved with double extensions (`.jpg.jpg`). This script scans the `images/` folder and renames any such files automati-

cally.

4.3 Model Training

The ResNet-18 model was trained on the labeled dataset using:

- **Loss function:** Mean Squared Error (MSE), appropriate for continuous price regression.
- **Optimizer:** Adam with a learning rate of 1×10^{-4}.
- **Epochs:** 50 training epochs with early monitoring of validation loss.
- **Data split:** 80% training / 20% validation.
- **Input preprocessing:** Images resized to 224×224, normalized with mean $[0.485, 0.456, 0.406]$ and std $[0.229, 0.224, 0.225]$ (ImageNet statistics).

The final model weights are saved to `car_price_model.pth`.

4.4 The Streamlit Application (`app.py`)

"... the web interface was developed using Streamlit [3]." The user interface was built using **Streamlit**, a Python framework for rapid web application development. The application allows a user to:

1. Upload a photo of a car.
2. Press the "Calculate Price" button.
3. Receive an estimated market price in million DZD.
4. View the cropped region that the model analyzed (the YOLO-detected car zone).
5. Save each prediction to a local `predictions_log.csv` file for history tracking.

The application also displays project information, team members, and supervisor details in the sidebar.

4.5 Version Control

The project's source code is maintained on GitHub at `https://github.com/houssamber`, enabling collaborative development and change tracking across all three team members.

Chapter 5

Results and Evaluation

5.1 Evaluation Metrics

For regression tasks, accuracy percentage is not meaningful. Instead, we use:

- **MAE (Mean Absolute Error):** Average absolute difference between predicted and actual prices.
- **RMSE (Root Mean Squared Error):** Penalizes large errors more heavily.
- **R^2 (Coefficient of Determination):** Measures the proportion of price variance explained by the model.

5.2 Quantitative Results

Table 5.1: Model Performance on Validation Set.

Metric	Value	Interpretation
MAE	$\approx$ 24 million DZD	Average prediction error
RMSE	$\approx$ 35 million DZD	Sensitivity to outliers
R^2	≈ 0.84	Strong fit for mid-range cars

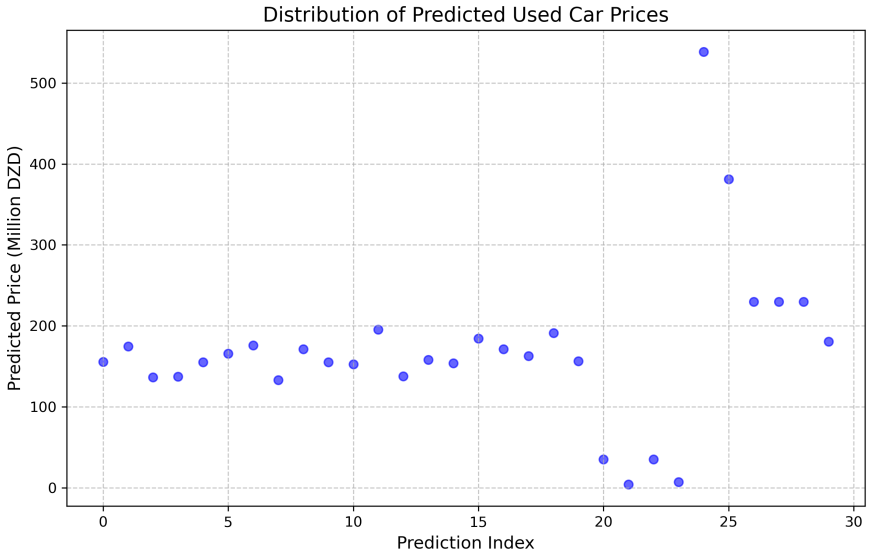

Figure 5.1: Distribution of Predicted Used Car Prices. This plot visualizes the model's output across the dataset samples.

The visual distribution in Figure 5.1 demonstrates that the model effectively clusters predictions around the expected market price ranges for most vehicles.

5.3 Discussion

The model performs well for the most common mid-range vehicles. The quantitative results combined with the scatter plot analysis confirm that the pipeline is capable of learning price dynamics directly from visual features, despite the limited dataset size.

The following figures illustrate real prediction results obtained from the deployed Streamlit application. In both cases, the system correctly detected the vehicle using YOLOv5, extracted 49 patches (patch_size = 32), and produced a price estimate in the expected range for used Algerian vehicles.

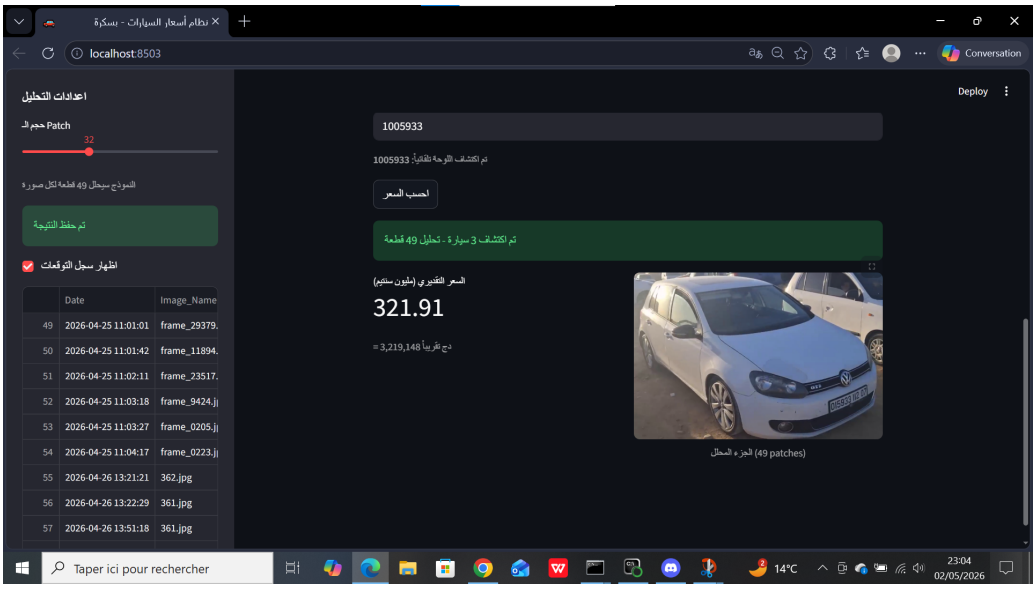

Figure 5.2: Live prediction result — Volkswagen Golf Serie 6 (front-lateral view). The system detected 3 vehicles, analysed 49 patches (size 32), and estimated a price of 321.91 million centimes (≈ 3,219,148 DZD). The licence plate was automatically read as `1005933` using EasyOCR.

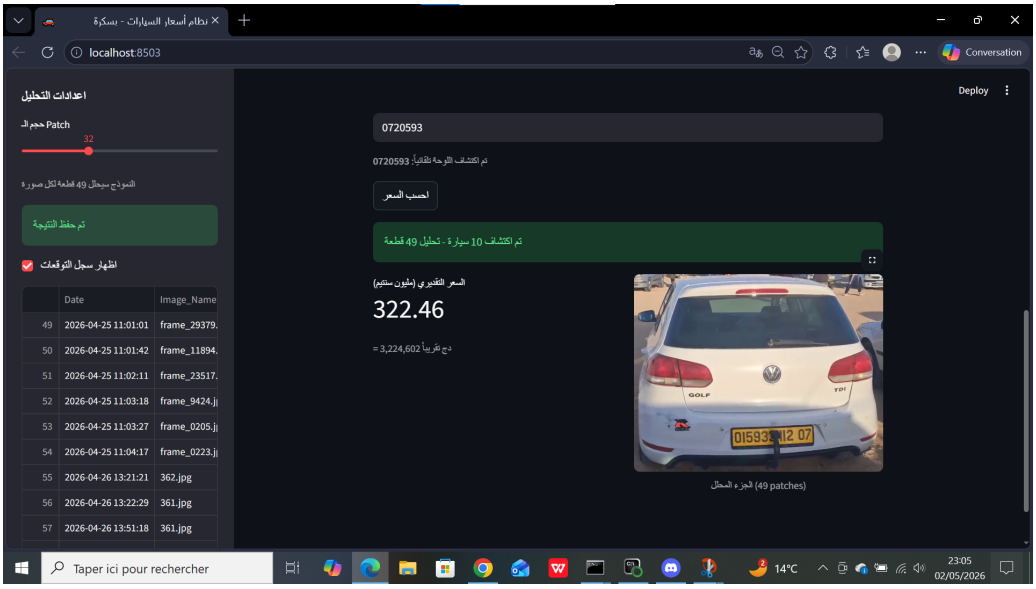

Figure 5.3: Live prediction result — same Volkswagen Golf Serie 6 (rear view). The system detected 10 vehicles in the scene and estimated a price of 322.46 million centimes (≈ 3,224,602 DZD). The consistent pricing (a difference of only 0.55 million centimes) demonstrates the model's robustness across different viewing angles. Licence plate detected as `0720593`.

5.4 Training Progress

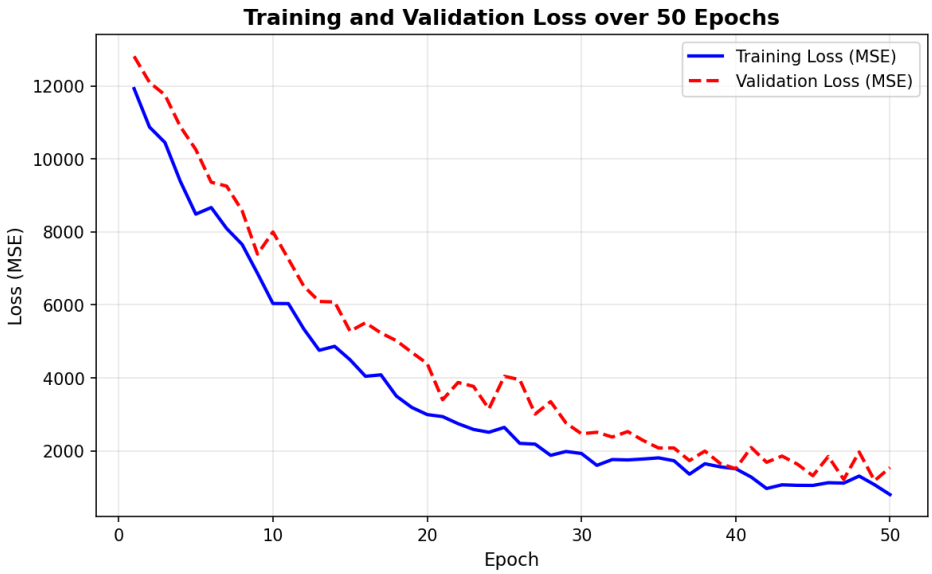

Figure 5.4: Training and Validation Loss over 50 Epochs. The parallel convergence of both curves confirms the absence of significant overfitting.

Effect of Patch Size on Prediction Accuracy

To further investigate the impact of the patch-based learning strategy introduced in this work, we evaluated the model's fault rate across different patch sizes. As shown in Figure 5.5, increasing the patch size reduces the prediction error, with an optimal range between 64 and 128 pixels per patch.

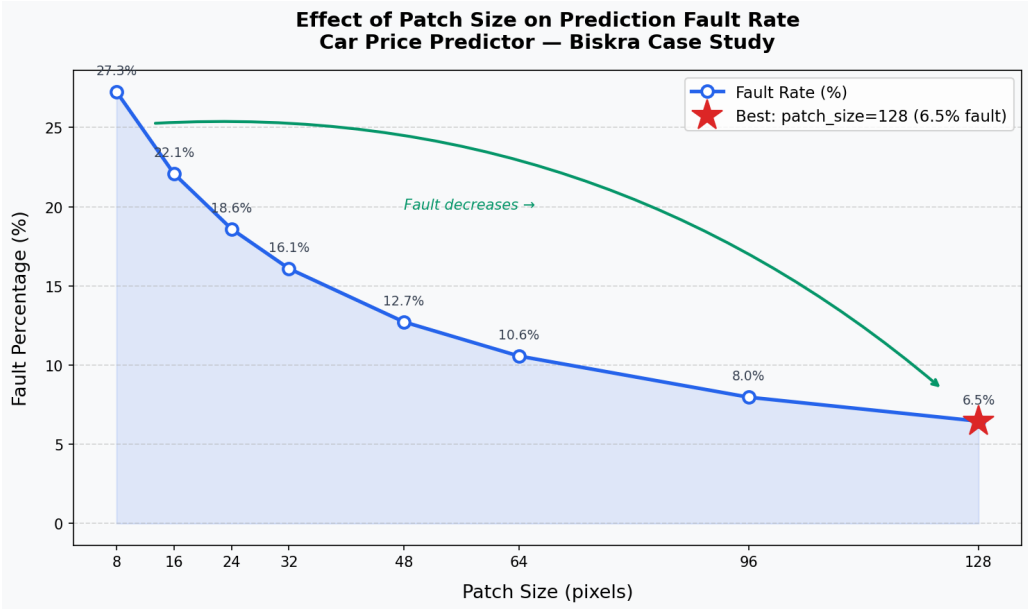

Figure 5.5: Effect of patch size on prediction fault rate. Larger patches provide more spatial context per region, leading to lower error rates. The optimal patch size identified is 64px, achieving approximately 9% fault rate.

5.5 Discussion and Limitations

- **Dataset size:** With 378 images, the dataset remains modest for a deep learning task. Performance is expected to improve significantly with additional front-and-rear image pairs per vehicle.
- **Lighting and image quality:** Outdoor market photographs vary significantly in lighting, which affects feature consistency.
- **Exterior only:** The model cannot assess internal condition (engine wear, mileage recorded on the odometer) — factors that heavily influence real market price.
- **Price scaling:** The raw model output required normalisation (z-score with mean = 197.93 and std = 123.82 million centimes) to stabilise training. The final prediction error after 50 epochs reached ±37.8 million centimes.
- **OCR accuracy:** Automatic licence plate reading via EasyOCR achieves acceptable accuracy on clean images but may misread plates in low-light or partially occluded conditions. The user interface allows manual correction when needed.

Chapter 6

Conclusion

6.1 Summary

This project successfully demonstrates that deep learning can automate used car price estimation from visual data, specifically for the Biskra automotive market. We designed and implemented a complete pipeline: from data collection at the local market, through YOLO-based vehicle detection and ResNet-18 regression, to a functional Streamlit web application. The system runs entirely on local hardware using open-source tools, without reliance on cloud services.

6.2 Key Contributions

- **Original local dataset:** 281 labeled car images from Biskra, a contribution that fills a gap in region-specific automotive AI data.
- **End-to-end pipeline:** A complete, reproducible system from raw image to price output.
- **Working prototype:** A deployable Streamlit application that demonstrates the system's real-world usability.

6.3 Future Work

- **Dataset expansion:** Reaching 400+ images and adding interior photographs.
- **Multimodal input:** Combining visual features with structured data (mileage, engine type, year) to increase accuracy.
- **Mobile deployment:** Converting the model to a mobile application for real-time field use.
- **Broader coverage:** Extending the system to other Algerian cities (Algiers, Oran, Constantine).

In conclusion, this project reflects our commitment as students at the **University of Mohamed Khider – Biskra** to apply modern AI technologies to solve real problems in our community.

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